DEX2013 Programming for Engineering

Python Assignment Report

Smart Machine Condition Monitoring and Safety Alert System

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1. Objective and dataset description

The objective of this assignment is to write a Python program that reads a set of simulated machine readings, decides whether each reading is **NORMAL**, **WARNING** or **CRITICAL**, calculates summary statistics, and plots the machine's behaviour over time. The program is a classroom decision-support exercise only. It does not connect to a real sensor and the thresholds are example values, not real equipment limits.

The dataset *machine_condition_dataset.csv* contains **30 readings** (rows) and **7 columns**, recorded every 10 seconds from 0 s to 290 s. Each row has a Reading_ID, the elapsed time (s), and five machine parameters: temperature (°C), pressure (bar), vibration (mm/s), motor current (A) and rotational speed (RPM). The file was loaded with pandas (`pd.read_csv`) and was not typed into the code.

2. Solution plan (problem statement, IPO and flowchart)

Problem statement. A maintenance team needs a simple program that checks every machine reading against fixed limits and reports which readings need attention, so that problems can be seen quickly and explained with numbers and graphs.

Inputs	Processing	Outputs
CSV file with Time_s, temperature, pressure, vibration, current and RPM for 30 readings	Load and check the CSV; classify each reading with classify_condition(); add Status and Recommended_Action; calculate mean, min and max; count and convert to percentages; plot graphs; run 10 tests	Status for every Reading_ID; statistics; counts and percentages; two PNG graphs; results CSV; test results

The flowchart below shows the decision for one reading. The **CRITICAL** check is made first. Only if no Critical condition is true does the program check for **WARNING**, and if neither applies the reading is **NORMAL**. A reading that meets both Critical and Warning conditions is therefore always classified as CRITICAL.

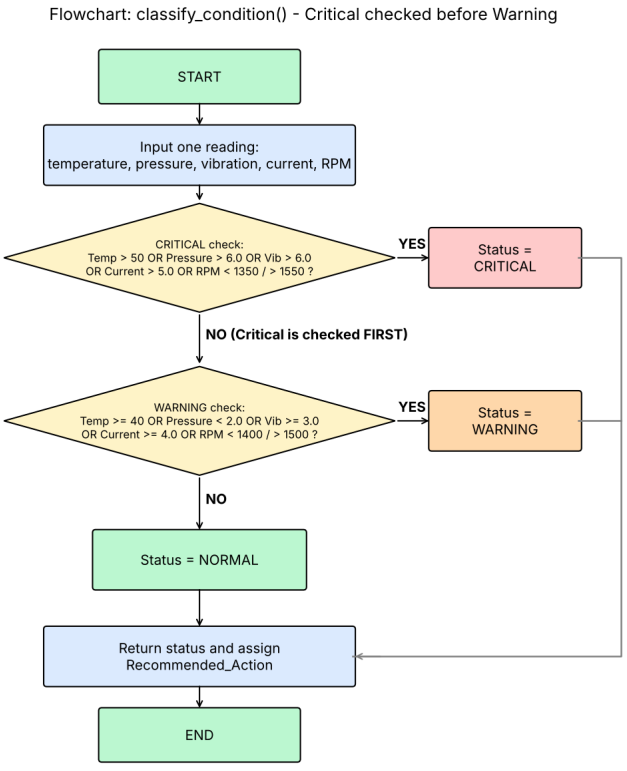


Figure 1: Flowchart for classifying one reading (Critical is checked before Warning).

3. Program output

The program was run in Python 3 with pandas and matplotlib. The screenshots below show the actual output: the CSV loading check, the status of every reading, the summary statistics, the status counts and percentages, and the saved graphs.

```
=== Smart Machine Condition Monitoring and Safety Alert System ===
(Classroom simulation only)

CSV file loaded successfully.
Number of readings (rows): 30
Number of columns: 7
All required columns are present.

First five rows:
   Reading_ID  Time_s  Temperature_C  Pressure_bar  Vibration_mm_s  Motor_Current_A   RPM
0           1       0           31.2           4.1             1.2              3.1  1480
1           2      10           32.0           4.0             1.3              3.2  1478
2           3      20           33.1           4.1             1.5              3.3  1482
3           4      30           34.0           3.9             1.7              3.4  1475
4           5      40           35.2           4.0             1.8              3.5  1470
```

Figure 2: CSV loaded successfully (30 rows, 7 columns) and the first five rows.

```
--- Reading status ---
 Reading_ID   Status
          1   NORMAL
          2   NORMAL
          3   NORMAL
          4   NORMAL
          5   NORMAL
          6   NORMAL
          7   NORMAL
          8   NORMAL
          9  WARNING
         10  WARNING
         11  WARNING
         12  WARNING
         13  WARNING
         14  WARNING
         15  WARNING
         16  WARNING
         17 CRITICAL
         18 CRITICAL
         19  WARNING
         20  WARNING
         21   NORMAL
         22 CRITICAL
         23 CRITICAL
         24 CRITICAL
         25 CRITICAL
         26 CRITICAL
         27  WARNING
         28  WARNING
         29  WARNING
         30   NORMAL
```

```
--- Summary statistics ---
Temperature_C:
  Mean = 39.44 °C
  Min  = 31.2 °C
  Max  = 52.5 °C
Pressure_bar:
  Mean = 3.56 bar
  Min  = 1.7 bar
  Max  = 6.2 bar
Vibration_mm_s:
  Mean = 3.21 mm/s
  Min  = 1.2 mm/s
  Max  = 6.5 mm/s
Motor_Current_A:
  Mean = 4.04 A
  Min  = 3.1 A
  Max  = 5.4 A
RPM:
  Mean = 1447.03 RPM
  Min  = 1320 RPM
  Max  = 1580 RPM

--- Status counts and percentages (total readings = 30) ---
NORMAL: 10 readings (33.3%)
WARNING: 13 readings (43.3%)
CRITICAL: 7 readings (23.3%)

Graphs saved: temperature_vs_time.png and vibration_vs_time.png

--- Classification tests ---
Test | Expected | Actual   | Result
1    | NORMAL   | NORMAL   | PASS
2    | WARNING  | WARNING  | PASS
3    | WARNING  | WARNING  | PASS
4    | CRITICAL | CRITICAL | PASS
5    | CRITICAL | CRITICAL | PASS
6    | CRITICAL | CRITICAL | PASS
7    | CRITICAL | CRITICAL | PASS
8    | CRITICAL | CRITICAL | PASS
9    | WARNING  | WARNING  | PASS
10   | WARNING  | WARNING  | PASS
10 of 10 tests passed.
```

Figure 3: Left – status of each Reading_ID. Right – summary statistics, status counts and percentages, saved graphs and the 10 test results.

4. Status counts and percentages

Status	Count	Percentage	Reading_IDs
NORMAL	10	33.3 %	1–8, 21, 30
WARNING	13	43.3 %	9–16, 19, 20, 27, 28, 29
CRITICAL	7	23.3 %	17, 18, 22, 23, 24, 25, 26
Total	30	100 %	

Parameter	Mean	Minimum	Maximum
Temperature (°C)	39.44	31.2	52.5
Pressure (bar)	3.56	1.7	6.2
Vibration (mm/s)	3.21	1.2	6.5
Motor current (A)	4.04	3.1	5.4
RPM	1447.03	1320	1580

Table 1: Summary statistics for the five machine parameters (30 readings).

5. Graphs

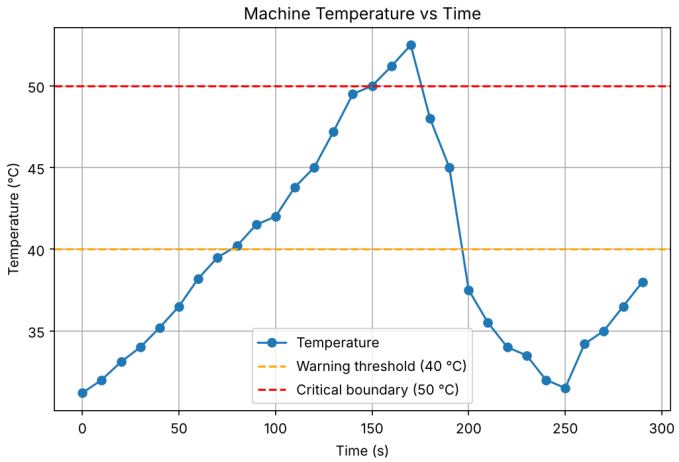


Figure 4: Temperature vs time with the Warning threshold (40 °C) and Critical boundary (50 °C).

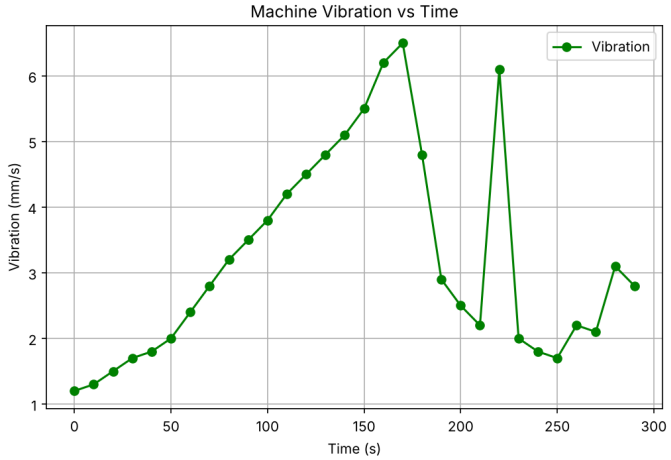


Figure 5: Vibration vs time.

6. Testing

The ten required test cases were run through classify_condition(). All ten returned the expected status, including the boundary cases (Test 3: 50 °C is not greater than 50, so WARNING; Test 4: 50.1 °C gives CRITICAL; Test 9: RPM 1390 is outside the normal band but not Critical).

Test	Temp °C	Pressure bar	Vib. mm/s	Current A	RPM	Expected	Actual	Result
1	35	4.0	2.0	3.5	1450	NORMAL	NORMAL	PASS
2	40	4.0	2.0	3.5	1450	WARNING	WARNING	PASS
3	50	4.0	2.0	3.5	1450	WARNING	WARNING	PASS
4	50.1	4.0	2.0	3.5	1450	CRITICAL	CRITICAL	PASS
5	35	6.1	2.0	3.5	1450	CRITICAL	CRITICAL	PASS
6	35	4.0	6.1	3.5	1450	CRITICAL	CRITICAL	PASS
7	35	4.0	2.0	5.1	1450	CRITICAL	CRITICAL	PASS
8	35	4.0	2.0	3.5	1349	CRITICAL	CRITICAL	PASS
9	35	4.0	2.0	3.5	1390	WARNING	WARNING	PASS
10	35	1.9	2.0	3.5	1450	WARNING	WARNING	PASS

Table 2: Required classification tests – expected and actual results (10 of 10 passed).

7. Engineering interpretation

Observation 1 – Temperature, vibration and current rise together to a peak. From 0 s to 170 s temperature rose from 31.2 °C to 52.5 °C, vibration from 1.2 to 6.5 mm/s and motor current from 3.1 to 5.4 A, while pressure fell steadily from 4.1 to 2.5 bar. The temperature and vibration graphs follow a similar rising shape and peak at the same time (Reading 18, 170 s), and their correlation in this dataset is about 0.84. Vibration changed the most in relative terms (its range is about 165 % of its mean). The first Warning appears at Reading 9 (80 s), where temperature (40.2 °C), vibration (3.2 mm/s) and current (4.0 A) all reach their Warning limits at once. Readings 17 and 18 are Critical because temperature, vibration and current all exceed their Critical limits together.

Observation 2 – Reading 16 sits exactly on a boundary. At 150 s the temperature is exactly 50.0 °C and the current is exactly 5.0 A. Because the Critical rules use "greater than", Reading 16 is classified as WARNING, not CRITICAL, even though it is on the edge. The next reading (17) is Critical, which shows how close the machine was to the Critical limits for several readings.

Observation 3 – Readings 22 to 26 are Critical for reasons other than temperature. After the machine cooled (temperature 31.5–35.5 °C), five consecutive readings were Critical for single, separate causes: Reading 22 pressure 6.2 bar, Reading 23 vibration 6.1 mm/s, Reading 24 current 5.3 A, Reading 25 RPM 1320 (below 1350) and Reading 26 RPM 1580 (above 1550). Each reading breaks a different single limit while the other parameters stay in range (the RPM swings from 1320 to 1580 and then 1390), so the temperature graph alone would not show these events. There are also low-pressure Warnings at Reading 19 (1.8 bar) and Reading 28 (1.7 bar).

8. Recommendation

Only 10 of 30 readings (33.3 %) were Normal, and 20 of 30 readings (66.7 %) were Warning or Critical, so in this simulation the machine should be inspected. The inspection should cover two things: (1) the heat and vibration build-up between about 80 s and 180 s, which crossed the Critical limits at 160–170 s; and (2) the separate pressure, vibration, current and RPM spikes at 210–250 s, which occurred while the temperature was normal. Readings 17, 18 and 22–26 would trigger a stop

and a supervisor or technician inspection under the classroom rules.

Before making a real operational decision more information would be needed: the machine type and its real manufacturer limits, the load and ambient temperature during the session, sensor calibration (to check whether the single-reading spikes are real events or sensor errors), the maintenance history, and a longer monitoring period to see whether the spikes repeat. This is a simulated dataset with example thresholds, so these results are for learning and do not replace a qualified engineer's assessment.